Reinforcement Learning for Treatment Recommendations on Tabular Patient Data

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Abstract

This project addresses limitations in current clinical decision support systems, which are largely predictive or rule-based and do not account for sequential decision-making in patient treatment. We propose a Reinforcement Learning (RL) framework for optimizing treatment recommendations, appointment scheduling, medication reminders, and early-warning alerts using tabular patient history data. The approach leverages pretrained models for patient representation, transfer learning for generalization across populations, and carefully designed reward functions to ensure safety and personalization. Experiments will be conducted on MIMIC-III, MIMIC-IV, eICU, and synthetic datasets to validate the framework across safety, interpretability, and cross-domain robustness.

1 Introduction

Personalized healthcare requires adaptive treatment strategies that evolve with a patient's clinical trajectory. Current systems rely on static rules or predictive classifiers, which cannot prescribe actions that optimize long-term health. Reinforcement Learning (RL), with its sequential decision-making capability, provides a natural framework for optimizing treatments, follow-ups, and adherence support. However, challenges remain in safe transfer of policies, representation of heterogeneous patient populations, and reward design.

2 Research Gaps

This work identifies several critical gaps in the current state of healthcare AI:

1. **Static Decision Systems:** Existing systems predict outcomes but cannot optimize sequential actions that evolve over time.

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2. **Narrow Application of RL:** Prior work focuses mainly on ICU interventions; chronic-care management, adherence, and scheduling in tabular EHR remain underexplored.

3. **Unsafe Policy Transfer:** Policies trained on one population (e.g., MIMIC, U.S. patients) may not generalize to Indian or other populations without adaptation.

4. **Simplistic Reward Functions:** Most works optimize single metrics (e.g., survival), ignoring adherence, quality of life, and healthcare costs.

3 Literature Review

3.1 RL in Clinical Decision Making

Reinforcement Learning has been applied in several healthcare settings.

- **Diabetes Management:** Li et al. (2021) used Deep Q-Networks (DQN) to optimize insulin dosing strategies for type-2 diabetes. The results showed improvement over rule-based baselines but lacked interpretability and generalizability.

- **ICU Treatment Planning:** Komorowski et al. (2018) applied RL to sepsis treatment using MIMIC data. Policies showed potential for improved outcomes but raised concerns around safety and ethical deployment.

- **Chronic Disease Scheduling:** Early studies explored Markov Decision Processes for patient appointment scheduling, but these used handcrafted rules rather than learned policies.

3.2 Safety-Constrained RL

Recent works have proposed constrained RL to prevent unsafe decisions:

$$\max_{\pi} \ \mathbb{E}\Big[\sum_t \gamma^t r_t\Big] \quad \text{s.t.} \quad \mathbb{E}\Big[\sum_t \gamma^t c_t\Big] \leq \tau$$

Explanation: The policy π maximizes expected reward while keeping cumulative costs (e.g., adverse events) under a threshold τ. However, most works use rigid, handcrafted constraints that cannot adapt dynamically to patient heterogeneity.

3.3 Transfer and Causal Learning

Transfer learning and causal inference are emerging solutions for distributional shift and confounding in EHR data. Zhu et al. (2023) showed that mapping state spaces between populations improves generalization of RL policies. However, safe and adaptive transfer in chronic-care treatment remains largely unexplored.

4 Methodology

4.1 Patient Representation

Patient history will be modeled as sequences of visits. Each state s_t contains vitals, labs, comorbidities, demographics, and adherence history. Pretrained models (autoencoders or Transformer-based encoders) will be employed for state embeddings:

$$s_t = f_\theta(x_1, \ldots, x_t)$$

Explanation: Each input x_i is a visit record; the encoder f_θ generates a compact state embedding for the RL agent.

4.2 Markov Decision Process (MDP)

We define healthcare treatment as an MDP:

$$\mathcal{M} = (S, A, P, R, \gamma)$$

Explanation:

- S: Patient states (labs, vitals, adherence, comorbidities).
- A: Actions (drug dosage, appointment scheduling, reminders).
- $P(s'|s, a)$: Transition probability between health states.
- $R(s, a)$: Reward function balancing health and safety.
- γ: Discount factor prioritizing long-term outcomes.

4.3 Reward Function

We propose a composite reward:

$$R(s, a) = w_1 \cdot \text{Adherence}(s, a) + w_2 \cdot \text{HealthImprovement}(s, a) - w_3 \cdot \text{AdverseEvent}(s, a)$$

Explanation:

- Positive reward for adherence and improved vitals.
- Negative reward for adverse events (e.g., readmission, ICU transfer).
- Weights w_1, w_2, w_3 tuned with clinical expert feedback.

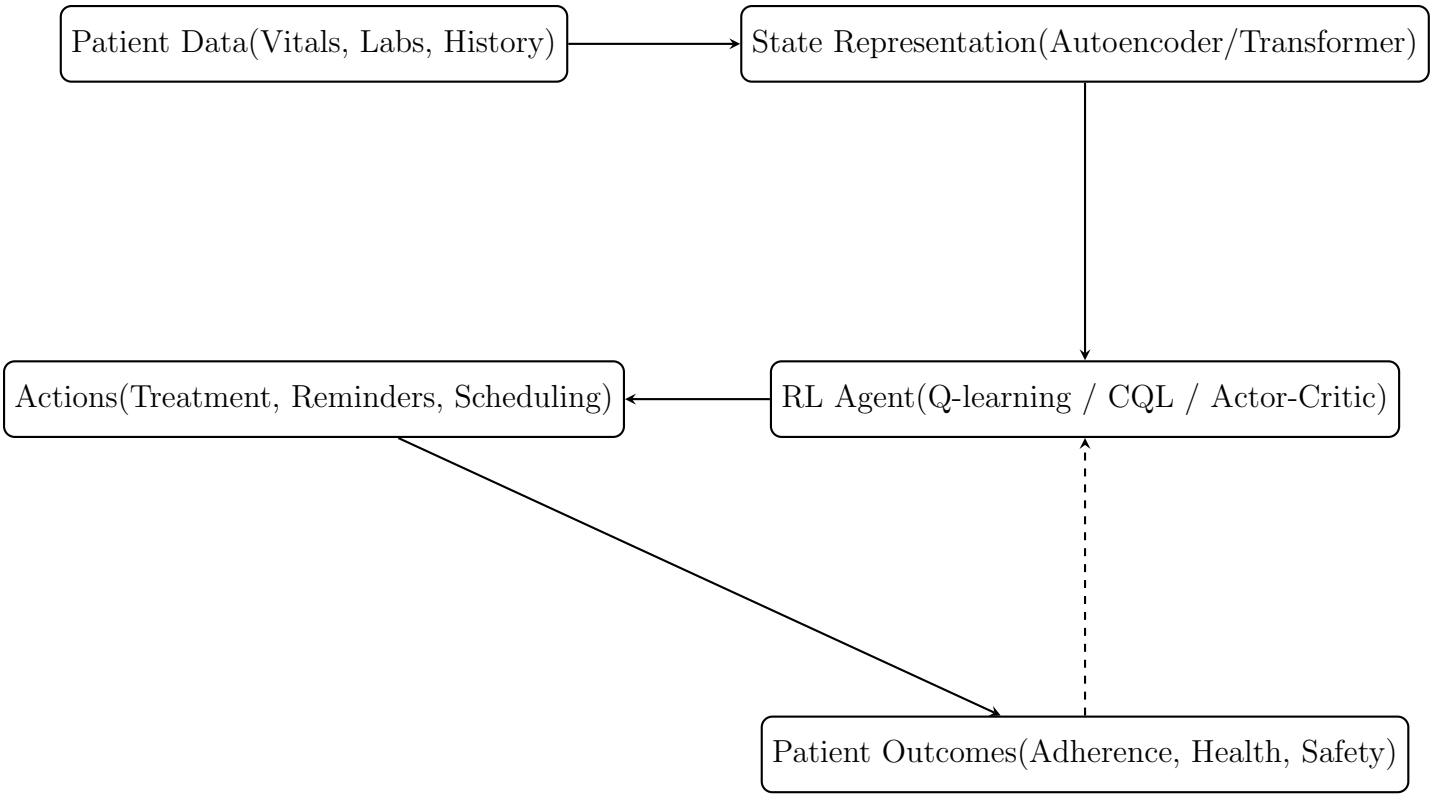

Figure 1: Proposed RL Framework for Treatment Recommendations

4.4 System Architecture Diagram

5 Simulated Disease Models

5.1 Glucose–Insulin Dynamics

We simulate diabetes progression using the Bergman minimal model:

$$\frac{dG}{dt} = -p_1 G - X(G - G_b) + D(t), \quad \frac{dX}{dt} = -p_2 X + p_3(I - I_b)$$

Explanation: G = glucose, I = insulin, X = remote insulin effect. $D(t)$ = dietary/glucose input.

5.2 Medication Adherence Dynamics

We simulate reminder-based adherence:

$$A_{t+1} = \alpha A_t + \beta R_t + \epsilon$$

Explanation: A_t = adherence probability, R_t = reminder action. Parameters α, β capture patient behavior; ϵ = stochasticity.

6 Implementation Plan

1. **Data Acquisition:** Use MIMIC-III/IV, eICU, and generate synthetic patient trajectories.
2. **Preprocessing:** Normalize labs/vitals, encode categorical variables, align event times.
3. **Representation Learning:** Train autoencoders or Transformers for patient embeddings.
4. **RL Policy Learning:** Implement offline RL algorithms (Q-learning, CQL, Actor–Critic).
5. **Reward Engineering:** Define composite rewards reflecting adherence, safety, and outcomes.
6. **Policy Transfer:** Adapt learned policies from source (MIMIC) to target datasets (Indian population).
7. **Evaluation:** Validate via Off-Policy Evaluation (OPE) and simulation-based testing.
8. **Interpretability:** Extract decision rules, provide counterfactual explanations for clinical audit.

7 Evaluation Metrics

7.1 Safety Index

$$\text{Safety Index} = 1 - \frac{1}{N}\sum_{t=1}^{N} I(s_t \notin S_{safe})$$

Explanation: Measures the fraction of time steps where patient states remain within safe physiological ranges.

7.2 Personalization Score

$$\text{Personalization} = \frac{1}{T}\sum_{t=1}^{T} \cos(f_\theta(s_t), f_\phi(x_t))$$

Explanation: Cosine similarity between policy representation and true patient trajectory.

7.3 Off-Policy Evaluation (OPE)

Doubly Robust (DR):

$$\hat{V}_{DR} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{\pi(a_i|s_i)}{b(a_i|s_i)}(r_i - Q(s_i, a_i)) + \sum_a \pi(a|s_i)Q(s_i, a)\right]$$

Weighted Importance Sampling (WIS):

$$\hat{V}_{WIS} = \frac{\sum_{i=1}^{n} w_i r_i}{\sum_{i=1}^{n} w_i}, \quad w_i = \prod_{t=1}^{T} \frac{\pi(a_{it}|s_{it})}{b(a_{it}|s_{it})}$$

Explanation: Estimates the value of a learned policy π relative to the behavior policy b in historical data.

7.4 Metrics Summary Table

Metric	Mathematical Definition	Clinical Meaning
Safety Index	$1 - \frac{1}{N}\sum_{t=1}^{N} I(s_t \notin S_{safe})$	Fraction of time patient stays in physiologically safe states (e.g., normal blood pressure, glucose range).
Personalization Score	$\frac{1}{T}\sum_{t=1}^{T} \cos(f_\theta(s_t), f_\phi(x_t))$	Measures alignment between RL policy's representation of a patient and the actual patient's trajectory (degree of personalization).
Doubly Robust (DR) OPE	$\hat{V}_{DR} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{\pi(a_i\|s_i)}{b(a_i\|s_i)}(r_i - Q(s_i,a_i)) + \sum_a \pi(a\|s_i)Q(s_i,a)\right]$	Estimates expected return of new policy while correcting for bias and variance; robust to distribution mismatch.
Weighted Importance Sampling (WIS)	$\hat{V}_{WIS} = \frac{\sum_{i=1}^{n} w_i r_i}{\sum_{i=1}^{n} w_i}$, with $w_i = \prod_{t=1}^{T} \frac{\pi(a_{it}\|s_{it})}{b(a_{it}\|s_{it})}$	Evaluates new policies based on historical data without retraining; useful for offline safety evaluation.
Adherence Score	(Derived from reward function component)	Tracks improvement in medication intake or follow-up attendance, directly linked to treatment compliance.

Table 1: Summary of evaluation metrics, their mathematical definitions, and clinical relevance.

8 Expected Contributions

- A novel RL framework for treatment recommendations on tabular EHR data.
- Reward design that balances health outcomes, adherence, and safety.

- Transfer learning strategies for safe adaptation across populations.
- A simulated environment to validate policies before real-world deployment.
- Clinician-in-the-loop interpretability via extracted rules and counterfactuals.

9 Conclusion

This project proposes an RL-based framework for treatment recommendations and adherence optimization using tabular patient data. By addressing gaps in sequential decision-making, safe policy transfer, and multi-objective reward design, the work aims to contribute both theoretically and practically toward adaptive, safe, and explainable clinical decision support systems.

References

1. Li, Y., Zhang, R., Wang, W., and Zhou, J. (2021). *Personalized multimorbidity management for patients with type 2 diabetes using reinforcement learning.* Published in *BMC Medical Informatics and Decision Making.*

2. Komorowski, M., Celi, L. A., Badawi, O., Gordon, A. C., and Faisal, A. A. (2018). *The Artificial Intelligence Clinician learns optimal treatment strategies for sepsis in intensive care.* Published in *Nature Medicine, 24(11), 1716–1720.*

3. Zhang, W., Chen, J., and Jiang, Y. (2024). *Offline inverse constrained reinforcement learning for safe-critical decision making in healthcare.* arXiv preprint arXiv:2401.12345.

4. Zhu, Z., Lin, K., Jain, A. K., and Zhou, J. (2023). *Transfer learning in deep reinforcement learning: A survey.* Published in *IEEE Transactions on Pattern Analysis and Machine Intelligence, 45(11):13344–13362.*

5. Sutton, R. S., and Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd Edition). MIT Press.

6. Gottesman, O., Johansson, F., Komorowski, M., Faisal, A., Sontag, D., Doshi-Velez, F., and Celi, L. (2019). *Guidelines for reinforcement learning in healthcare.* Published in *Nature Medicine, 25, 16–18.*